

Notes: Aguirregabiria and Mira (Econometrica, 2007)

Sequential Estimation of Dynamic Discrete Games

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1 Big picture

- **Question.** How can we estimate dynamic discrete games of incomplete information when the model may have **multiple equilibria** and solving the game repeatedly is computationally expensive?
- **Setting.** The paper studies a class of dynamic oligopoly games in which players make discrete decisions over time, observe common state variables, and also receive private payoff shocks.
- **Core challenge.** Standard maximum likelihood is hard for two reasons. First, with multiple equilibria the model does not deliver a unique reduced form for each parameter vector. Second, every evaluation of the likelihood may require solving a dynamic game again, which becomes extremely costly as the number of players and state variables grows.
- **Main methodological result.** The paper shows that Markov perfect equilibria can be represented as fixed points of a **best-response mapping in the space of choice probabilities**. This makes it possible to build pseudo maximum likelihood estimators that do not require solving for all equilibria at every trial parameter value.
- **Main econometric result.** The paper develops two estimators. A **two-step pseudo maximum likelihood (PML)** estimator is simple and consistent if the researcher has a good first-step estimate of players' choice probabilities. A recursive extension, the **nested pseudo likelihood (NPL)** estimator, iterates between parameter estimation and probability updating, reducing finite-sample bias and allowing estimation even when a consistent first-step nonparametric estimator is unavailable.
- **Main quantitative result.** In Monte Carlo experiments, the two-step estimator based on raw frequency estimates has very large small-sample bias, while NPL performs much better and is often close to the infeasible benchmark that knows the true choice probabilities.
- **Main empirical result.** In Chilean local retail industries, the model delivers sensible estimates of fixed operating costs, sunk entry costs, strategic interactions, and unobserved market heterogeneity. Restaurants have relatively low fixed costs and weak strategic interactions; gas stations have the largest sunk entry costs; bookstores combine high fixed costs with weak strategic interaction.
- **Why the paper matters.** This paper is a benchmark for empirical dynamic games because it shows how to estimate them without imposing equilibrium uniqueness everywhere and without paying the full computational cost of nested fixed-point maximum likelihood.

2 Data and measurement (key objects)

2.1 Observed market-level panel in the econometric framework

- **Sampling framework.** The researcher observes many geographically separate markets over a short panel. The data are

$$\text{data} = \{a_{mt}, x_{mt} : m = 1, \dots, M; t = 1, \dots, T\},$$

where M is large and T is small.

- **Observed outcomes.** For each market and date, the econometrician observes the vector of players' actions a_{mt} and the vector of common-knowledge state variables x_{mt} .
- **Global versus local players.** The framework allows some players to appear in many markets (global players) and others to appear only in one local market (local players). This distinction matters because player heterogeneity can only be estimated for players who appear in enough markets.
- **Typical empirical use.** This is the standard industrial-organization environment with many local markets, such as entry and exit of retailers, banks, or service stations across towns.

2.2 States, unobservables, and timing in the model

- **Common state variables.** The vector x_t contains payoff-relevant variables that are common knowledge, such as market size and lagged incumbency status.
- **Private information.** Each player i also receives a private payoff shock ε_{it} observed only by that player.
- **Key assumptions.** The paper assumes additive separability of private shocks in profits, conditional independence of common and private states, independent private values across players, and a discrete finite support for common states.
- **Timing.** At the beginning of each period, firms observe the common state and their own private shock, choose actions simultaneously, earn current profits, and then the common state transitions according to a controlled Markov process.
- **Economic interpretation.** Players understand both the law of motion of the observable state and the strategic environment, but each player has its own private current-profit realization.

2.3 Monte Carlo design used to evaluate the estimators

- **Model.** The Monte Carlo section studies a dynamic entry-exit game with $N = 5$ global players, binary activity choices, and a market-size state with five points of support.
- **Profit function.** Variable profits are log-linear in market size and in the number of competitors, with additional fixed operating costs, sunk entry costs, and private shocks.
- **State space size.** The common state (S_{mt}, a_{t-1}) has $5 \times 2^5 = 160$ points, and the full vector of conditional choice probabilities has dimension $5 \times 160 = 800$.
- **Sample size.** The reported simulations use $M = 400$ markets and 1,000 Monte Carlo replications.
- **Purpose.** The simulations are designed to isolate how estimator performance changes with strategic interaction and sunk entry costs.

2.4 Empirical application: Chilean local retail markets

- **Source.** The empirical application uses a Chilean tax census from the Servicio de Impuestos Internos, which covers all firms paying the sales tax.

- **Period.** The firm-level panel has annual frequency and covers 1994–1999. Population data at the comuna level are added for 1990–2003.
- **Firm-level variables.** The data contain firm ID, five-digit industry, annual sales (grouped into 12 cells), and comuna location.
- **Markets.** A local market is a comuna. After excluding metropolitan areas and very large comunas, the working sample contains 189 comunas.
- **Industries.** The paper estimates separate models for restaurants, gas stations, bookstores, shoe shops, and fish shops.
- **Descriptive facts.** Restaurants are much more numerous than the other retailers, concentration is lower in restaurants, turnover is high in all industries, and gas stations exhibit the highest short-run survival.

3 Models and empirical specifications (all equations + interpretation)

3.1 Dynamic discrete game and current-profit specification

There are N players, each choosing an action from a finite set $A = \{0, 1, \dots, J\}$. The common state is x_t , private information is $\varepsilon_t = (\varepsilon_{1t}, \dots, \varepsilon_{Nt})$, and current profits are

$$\tilde{\Pi}_i(a_t, x_t, \varepsilon_{it}) = \Pi_i(a_t, x_t) + \varepsilon_{it}(a_{it}).$$

Players maximize expected discounted profits,

$$\mathbb{E} \left[\sum_{s=t}^{\infty} \beta^{s-t} \tilde{\Pi}_i(a_s, x_s, \varepsilon_{is}) \mid x_t, \varepsilon_{it} \right].$$

- **Interpretation.** Private information affects only current payoff differences across actions, while the common state captures all persistent strategic information.
- **Why this matters.** The additive private-value structure is what allows the paper to move from strategies in payoff space to a fixed-point problem in probability space.

3.2 Strategies, conditional choice probabilities, and Bellman equations

For a strategy profile $\sigma = \{\sigma_i(x, \varepsilon_i)\}$, the conditional choice probability is

$$P_i^\sigma(a_i \mid x) = \Pr(\sigma_i(x, \varepsilon_i) = a_i \mid x).$$

Expected current profits given opponents' strategies are

$$\pi_i^\sigma(a_i, x) = \sum_{a_{-i} \in A^{N-1}} \left(\prod_{j \neq i} P_j^\sigma(a_{-i}[j] \mid x) \right) \Pi_i(a_i, a_{-i}, x).$$

The integrated Bellman equation is

$$V_i^\sigma(x) = \int \max_{a_i \in A} \{v_i^\sigma(a_i, x) + \varepsilon_i(a_i)\} g_i(d\varepsilon_i),$$

with choice-specific value

$$v_i^\sigma(a_i, x) = \pi_i^\sigma(a_i, x) + \beta \sum_{x' \in X} V_i^\sigma(x') f_i^\sigma(x' | x, a_i).$$

- **Interpretation.** Once opponents' behavior is summarized by choice probabilities, each player's dynamic problem becomes a standard discrete-choice dynamic program.
- **Why the CCP representation matters.** The model's strategic content gets compressed into conditional choice probabilities, which are easier to work with than full strategy functions.

3.3 Markov perfect equilibrium in probability space

A stationary Markov perfect equilibrium (MPE) requires each player's strategy to be a best response to others. In probability space, equilibrium probabilities P^* satisfy

$$P^* = \Lambda(P^*),$$

where each component $\Lambda_i(a_i | x; P_{-i})$ is the probability that action a_i maximizes the choice-specific value plus private shock.

- **Interpretation.** Equilibrium is a fixed point of players' conditional choice probabilities.
- **Key difficulty.** In general, this fixed point is not unique. Multiple equilibria are natural in these games.

3.4 Alternative best-response mapping and the representation lemma

The paper's main computational insight is to replace the full dynamic-programming best-response mapping with a simpler mapping in probability space. Define the valuation operator

$$\Gamma_i(P) = (I - \beta F^P)^{-1} \sum_{a_i \in A} P_i(a_i) * [\pi_i^P(a_i) + e_i^P(a_i)],$$

where $e_i^P(a_i)$ is the expected private shock conditional on action a_i being optimal.

Then the alternative best-response mapping is

$$\Psi_i(a_i | x; P) = \int I \left\{ a_i = \arg \max_{a \in A} \left[\pi_i^P(a, x) + \varepsilon_i(a) + \beta \sum_{x' \in X} \Gamma_i(x'; P) f_i^P(x' | x, a) \right] \right\} g_i(\varepsilon_i) d\varepsilon_i.$$

The **representation lemma** states that the fixed points of Λ and Ψ coincide.

- **Interpretation.** Instead of resolving a dynamic programming problem every time we compute a best response, we solve a linear system for valuations and then update probabilities.
- **Computational gain.** This is the main trick of the paper. It preserves the equilibrium set but is much cheaper to evaluate.

3.5 Binary entry-exit special case

In the binary entry-exit example, a firm's profit when active can be written as

$$\tilde{\Pi}_{imt}(1) = \theta_{RS} \ln(S_{mt}) - \theta_{RN} \ln\left(1 + \sum_{j \neq i} a_{jmt}\right) - \theta_{FC,i} - \theta_{EC}(1 - a_{i,m,t-1}) + \varepsilon_{imt},$$

and zero otherwise after normalization.

Under the separable structure of the paper, the best response can be written as

$$\Psi_i(1 \mid x_t; P) = G\left(\tilde{z}_i^P(x_t) \frac{\theta_\pi}{\sigma} + \tilde{\lambda}_i^P(x_t)\right),$$

where $G(\cdot)$ is the relevant CDF implied by the private-shock distribution.

- **Interpretation.** In the binary case, the dynamic game becomes a nonlinear probability model whose “regressors” are themselves equilibrium objects built from P .
- **Why this matters empirically.** This is what lets the paper estimate entry-exit games with standard pseudo-likelihood tools once P is treated properly.

3.6 The pseudo likelihood and why full MLE is hard

The paper defines the pseudo likelihood

$$Q_M(\theta, P) = \frac{1}{M} \sum_{m=1}^M \sum_{t=1}^T \sum_{i=1}^N \ln \Psi_i(a_{imt} \mid x_{mt}; P, \theta).$$

If equilibrium were unique, full MLE would solve

$$\hat{\theta}_{MLE} = \arg \max_{\theta \in \Theta} Q_M(\theta, P) \quad \text{subject to} \quad P = \Psi(\theta, P).$$

With multiple equilibria, one would additionally need to search over all fixed points P compatible with a given θ .

- **Interpretation.** The pseudo likelihood uses best-response probabilities evaluated at an arbitrary P . It becomes the true likelihood only when P is an equilibrium associated with θ .
- **Why full MLE is impractical.** For each trial θ , the researcher may need to compute all equilibria. This is exactly the problem the paper wants to avoid.

3.7 Two-step pseudo maximum likelihood

Suppose the population conditional choice probabilities P^0 were known. Then the infeasible PML estimator solves

$$\hat{\theta} = \arg \max_{\theta \in \Theta} Q_M(\theta, P^0).$$

If we replace P^0 with a root- M consistent first-step estimator $\hat{P}^0$, we obtain the feasible two-step estimator

$$\hat{\theta}_{2S} = \arg \max_{\theta \in \Theta} Q_M(\theta, \hat{P}^0).$$

Its asymptotic variance is

$$V_{2S} = \Omega_{\theta\theta}^{-1} + \Omega_{\theta\theta}^{-1} \Omega_{\theta P} \Sigma \Omega_{\theta P}' \Omega_{\theta\theta}^{-1},$$

where Σ is the variance of the first-step estimate of P^0 .

- **Interpretation.** The estimator is easy because P is treated as fixed in the second step.
- **Main weakness.** Any noise in the first step feeds directly into second-step bias and inefficiency. In high-dimensional state spaces, that is a serious problem.

3.8 Nested pseudo likelihood (NPL)

Starting from any initial guess $\hat{P}^0$, NPL iterates

$$\hat{\theta}^K = \arg \max_{\theta \in \Theta} Q_M(\theta, \hat{P}^{K-1}), \quad \hat{P}^K = \Psi(\hat{\theta}^K, \hat{P}^{K-1}).$$

Define

$$\phi_M(P) = \Psi(\tilde{\theta}_M(P), P), \quad \tilde{\theta}_M(P) = \arg \max_{\theta \in \Theta} Q_M(\theta, P).$$

A NPL fixed point satisfies both

$$P = \phi_M(P) \quad \text{and} \quad \theta = \tilde{\theta}_M(P).$$

The paper defines the NPL estimator as the fixed point with the highest pseudo-likelihood value.

- **Interpretation.** NPL alternates between “estimate the structural parameters given beliefs” and “update beliefs using the model.”
- **Why it is better.** It preserves the low computational cost of PML, reduces finite-sample bias, and does not require the initial guess for P to be consistent.
- **Efficiency result.** Under conditions on the Jacobian $\nabla_P \Psi$, the NPL estimator can be more efficient than both the two-step estimator and the infeasible PML benchmark.

3.9 Permanent unobserved heterogeneity and initial conditions

To handle permanent unobserved market heterogeneity, the paper lets profits include a market-specific effect ω_m ,

$$\tilde{\Pi}_{imt}(1) = \theta_{RS} \ln(S_{mt}) - \theta_{RN} \ln \left(1 + \sum_{j \neq i} a_{jmt} \right) - \theta_{FC,i} - \theta_{EC}(1 - a_{i,m,t-1}) + \omega_m + \varepsilon_{imt},$$

where ω_m has discrete support known up to parameters.

The likelihood integrates over the unobserved types, and the pseudo likelihood uses the steady-state distribution of the initial state to handle the initial-conditions problem:

$$Q_M(\theta, P, f) = \frac{1}{M} \sum_{m=1}^M \ln \left(\sum_{l=1}^L \phi_l(x_m) \prod_{t=1}^T \prod_{i=1}^N \Psi_i(a_{imt} \mid x_{mt}, \omega_l; P_l, \theta) p^*(x_{m1} \mid f, P_l) \right).$$

- **Interpretation.** Different observationally similar markets may actually belong to different latent types and may even play different equilibria.
- **Why NPL is useful here.** A consistent nonparametric estimate of choice probabilities is generally unavailable with permanent unobserved heterogeneity, but NPL still works.

3.10 One-step maximum likelihood refinement

After obtaining consistent estimates of θ^0 and P^0 from NPL, the paper shows how to build a one-step MLE using estimated scores and information matrices.

- **Interpretation.** NPL is the workhorse estimator, but one can use it as a launch point for a more efficient likelihood-based update.
- **Practical message.** The paper’s main contribution is not the final one-step refinement; it is the tractable route to a consistent and well-behaved initial structural estimator.

4 Identification and instruments (what makes the estimates credible)

4.1 What fails in the naive approach

- **Full MLE is not practically available.** With multiple equilibria, one cannot just write down a unique likelihood indexed by θ .
- **The computational problem is structural.** In dynamic games, best responses themselves come from dynamic programming problems, so solving the model inside every likelihood evaluation is very costly.
- **Nonparametric first steps can be too noisy.** In the two-step estimator, raw frequency estimates of choice probabilities can be extremely imprecise when the state space is large relative to the sample.

4.2 What identifies the structural parameters in the new approach

- **Single equilibrium in the observed subsample.** Assumption 5 states that the data are generated by one equilibrium P^0 for each observed market type.
- **Correct beliefs about future play.** Players are assumed to expect the same equilibrium object P^0 to govern future periods, which makes the dynamic continuation values well defined.
- **Structural uniqueness.** Identification requires that no $\theta \neq \theta_0$ together with an equilibrium P reproduces the same observed conditional choice probabilities P^0 .

- **Key implication.** The paper identifies the model through equilibrium conditional choice probabilities rather than through uniqueness of equilibrium as a function of parameters.

4.3 Why the representation lemma is central

- **Equivalent equilibrium sets.** The lemma says that the fixed points of the easy-to-compute mapping Ψ are exactly the equilibria of the original game.
- **This is the paper’s real computational identification device.** It lets the econometrician work with an object that is uniquely defined for any (θ, P) even though equilibrium probabilities themselves may not be unique functions of θ .

4.4 Why NPL improves on the two-step estimator

- **Bias correction through iteration.** NPL repeatedly projects noisy first-step probabilities back toward model-consistent probabilities.
- **Less reliance on nonparametrics.** Consistency of NPL does not require the initial guess $\hat{P}^0$ to be consistent.
- **Efficiency.** When strategic feedbacks are not too explosive, imposing the equilibrium restriction in the sample can improve efficiency.

4.5 Identification with unobserved heterogeneity

- **Finite latent types.** The unobserved market effect is assumed to have discrete finite support.
- **Known number of types.** The econometrician must know the number of latent market types for the likelihood mixture to be interpretable.
- **Initial conditions.** The initial state is handled through the stationary distribution implied by the model and the equilibrium probabilities for each latent type.
- **Interpretation.** The framework can even accommodate data generated by multiple equilibria across latent types, as long as the econometrician models those types correctly.

4.6 Relation to the literature

- The paper extends the pseudo-likelihood logic used in single-agent dynamic models by Hotz and Miller and by Aguirregabiria and Mira to dynamic games.
- It complements Pakes, Ostrovsky, and Berry’s two-step moment approach and Pesendorfer and Schmidt-Dengler’s minimum-distance perspective.
- Relative to static incomplete-information game estimators such as Tamer or Ciliberto and Tamer, the contribution here is to bring tractable pseudo-likelihood estimation into a genuinely dynamic strategic setting.

5 Tables and figures

5.1 Table I and Table II: Monte Carlo parameterization and equilibrium properties

TABLE I
MONTE CARLO EXPERIMENTS: PARAMETERS θ_{EC} AND θ_{RN}

Parameter ^a	Exp. 1 $\theta_{EC} = 1.0$ $\theta_{RN} = 0.0$	Exp. 2 $\theta_{EC} = 1.0$ $\theta_{RN} = 1.0$	Exp. 3 $\theta_{EC} = 1.0$ $\theta_{RN} = 2.0$	Exp. 4 $\theta_{EC} = 0.0$ $\theta_{RN} = 1.0$	Exp. 5 $\theta_{EC} = 2.0$ $\theta_{RN} = 1.0$	Exp. 6 $\theta_{EC} = 4.0$ $\theta_{RN} = 1.0$
θ_{EC}	0.91	0.91	0.91	0.00	1.82	3.64
$\theta_{RN} \ln(3)$						
$100 \frac{\theta_{RN} \ln(2)}{\theta_{RN} \ln(3)}$	0.0%	63.1%	126.2%	63.1%	63.1%	63.1%

^aThe parameter $\theta_{EC}/(\theta_{RN} \ln(3))$ represents the ratio between entry costs and the annual variable profit of a monopolist in a market of average size, i.e., $S_{mI} = 3$. The parameter $100(\theta_{RN} \ln(2))/(\theta_{RN} \ln(3))$ represents the percentage reduction in annual variable profits when we go from a monopoly to a duopoly in an average size market, i.e., $S_{mI} = 3$.

TABLE II
MONTE CARLO EXPERIMENTS: DESCRIPTION OF THE MARKOV PERFECT
EQUILIBRIUM IN THE DGP^a

Descriptive Statistics	Exp. 1 $\theta_{EC} = 1.0$ $\theta_{RN} = 0.0$	Exp. 2 $\theta_{EC} = 1.0$ $\theta_{RN} = 1.0$	Exp. 3 $\theta_{EC} = 1.0$ $\theta_{RN} = 2.0$	Exp. 4 $\theta_{EC} = 0.0$ $\theta_{RN} = 1.0$	Exp. 5 $\theta_{EC} = 2.0$ $\theta_{RN} = 1.0$	Exp. 6 $\theta_{EC} = 4.0$ $\theta_{RN} = 1.0$
Number of active firms						
Mean	3.676	2.760	1.979	2.729	2.790	2.801
Std. dev.	1.551	1.661	1.426	1.515	1.777	1.905
AR(1) for number of active firms (autoregressive parameter)	0.744	0.709	0.571	0.529	0.818	0.924
Number of entrants (or exits) per period	0.520	0.702	0.748	0.991	0.463	0.206
Excess turnover ^b (in # of firms per period)	0.326	0.470	0.516	0.868	0.211	0.029
Correlation between entries and exits	-0.015	-0.169	-0.220	-0.225	-0.140	-0.110
Prob. of being active						
Firm 1	0.699	0.496	0.319	0.508	0.487	0.455
Firm 2	0.718	0.527	0.356	0.523	0.521	0.501
Firm 3	0.735	0.548	0.397	0.547	0.556	0.550
Firm 4	0.753	0.581	0.434	0.564	0.592	0.610
Firm 5	0.770	0.607	0.475	0.586	0.632	0.686

^aFor all these experiments, the values of the rest of the parameters are $N = 5$, $\theta_{FC,1} = -1.9$, $\theta_{FC,2} = -1.8$, $\theta_{FC,3} = -1.7$, $\theta_{FC,4} = -1.6$, $\theta_{FC,5} = -1.5$, $\theta_{RS} = 1.0$, $\sigma_e = 1$, and $\beta = 0.95$.

^bExcess turnover is defined as $(\#entrants + \#exits) - \text{abs}(\#Entrants - \#Exits)$.

Read:

- Table I varies two key primitives across experiments: **sunk entry costs** θ_{EC} and **strategic interaction** θ_{RN} .
- Table II shows the induced equilibrium objects. Stronger strategic interaction lowers the average number of active firms and raises turnover. Higher entry costs reduce turnover and make market structure more persistent.
- This pair of tables tells you what the Monte Carlo is trying to stress-test: whether the estimators behave worse when the game is more strategic or when entry dynamics are more important.

5.2 Table III, Table IV, and Table V: finite-sample performance of two-step PML and NPL

Read:

- Table III reports the median number of NPL iterations. Convergence is slower when strategic interactions are stronger, but the algorithm still converges reliably in the reported experiments.
- Table IV is the core finite-sample table. The two-step estimator based on frequency estimates is severely biased in many experiments, especially for fixed costs and the market-size coefficient. NPL stays close to the true values and close to the infeasible benchmark that knows the true choice probabilities.
- Table V shows the same point using relative root mean squared error. NPL is never far from the infeasible benchmark and can even dominate it in some experiments with strong strategic interaction.

TABLE IV
MONTE CARLO EXPERIMENTS: EMPIRICAL MEANS AND EMPIRICAL STANDARD
DEVIATIONS OF ESTIMATORS

Experiment	Estimator	Parameters			
		$\theta_{C,1}$	θ_{RS}	θ_{EC}	θ_{EN}
1	True values	-1.900	1.000	1.000	0.000
	2S-True	-1.915 (0.273)	1.007 (0.132)	1.002 (0.139)	0.002 (0.422)
	2S-Freq	-0.458 (0.289)	0.374 (0.141)	1.135 (0.190)	0.200 (0.364)
	2S-Logit	-1.929 (0.279)	1.006 (0.153)	0.997 (0.138)	-0.009 (0.431)
	NPL	-1.902 (0.279)	1.018 (0.157)	0.994 (0.139)	0.036 (0.439)
2	True values	-1.900	1.000	1.000	1.000
	2S-True	-1.894 (0.212)	1.002 (0.186)	1.007 (0.118)	1.007 (0.583)
	2S-Freq	-0.919 (0.208)	0.351 (0.119)	0.886 (0.125)	0.095 (0.337)
	2S-Logit	-1.920 (0.226)	0.977 (0.197)	1.000 (0.122)	0.915 (0.597)
	NPL	-1.893 (0.232)	1.016 (0.220)	0.998 (0.121)	1.050 (0.681)
3	True values	-1.900	1.000	1.000	2.000
	2S-True	-1.910 (0.183)	1.006 (0.209)	1.000 (0.112)	2.008 (0.783)
	2S-Freq	-1.126 (0.180)	0.286 (0.084)	0.792 (0.107)	0.027 (0.311)
	2S-Logit	-1.919 (0.248)	1.022 (0.305)	0.985 (0.145)	2.070 (1.110)
	NPL	-1.920 (0.232)	0.950 (0.189)	1.007 (0.116)	1.792 (0.667)
4	True values	-1.900	1.000	0.000	1.000
	2S-True	-1.900 (0.516)	1.020 (0.329)	0.001 (0.119)	1.063 (1.345)
	2S-Freq	-0.910 (0.243)	0.337 (0.104)	0.239 (0.113)	0.127 (0.354)
	2S-Logit	-2.070 (0.456)	0.903 (0.262)	0.000 (0.119)	0.571 (1.061)
	NPL	-1.891 (0.482)	1.014 (0.291)	0.001 (0.115)	1.047 (1.186)
5	True values	-1.900	1.000	2.000	1.000
	2S-True	-1.912 (0.178)	1.007 (0.142)	2.008 (0.132)	1.006 (0.359)
	2S-Freq	-0.840 (0.218)	1.379 (0.150)	1.591 (0.145)	0.181 (0.302)
	2S-Logit	-1.921 (0.204)	0.997 (0.167)	2.002 (0.138)	0.971 (0.405)
	NPL	-1.924 (0.203)	1.018 (0.178)	2.000 (0.137)	1.027 (0.435)
6	True values	-1.900	1.000	4.000	1.000
	2S-True	-1.899 (0.206)	1.003 (0.132)	4.050 (0.203)	1.006 (0.238)
	2S-Freq	-0.558 (0.228)	0.332 (0.128)	2.745 (0.211)	0.206 (0.238)
	2S-Logit	-1.895 (0.240)	0.996 (0.147)	4.048 (0.208)	0.992 (0.277)
	NPL	-1.918 (0.239)	1.009 (0.152)	4.044 (0.207)	1.009 (0.285)

TABLE III
MONTE CARLO EXPERIMENTS: MEDIAN NUMBER
OF ITERATIONS OF THE NPL ALGORITHM

Experiment	Initial Probabilities		
	Frequencies	Logits	Random
1	8	4	6
2	11	7	9
3	27	19	23
4	16	8	11
5	12	7	9
6	13	9	10

TABLE V
SQUARE-ROOT MEAN SQUARE ERROR RELATIVE TO THE ONE-STAGE PML WITH TRUE θ^0

Experiment	Estimator	Parameters			
		$\theta_{C,1}$	θ_{RS}	θ_{EC}	θ_{EN}
1	2S-Freq	5.380	4.222	1.676	0.983
	2S-Logit	1.027	1.006	1.002	1.022
	NPL	1.019	1.040	0.996	1.044
	2S-True	4.736	3.553	1.415	1.655
2	2S-Freq	1.070	1.066	1.029	1.034
	2S-Logit	1.098	1.188	1.020	1.171
	NPL	4.347	3.440	2.095	2.549
	2S-True	1.357	1.462	1.301	1.419
3	2S-Freq	1.268	0.935	1.038	0.892
	2S-Logit	1.977	2.035	2.228	0.699
	NPL	0.906	0.848	1.000	0.850
	2S-True	0.955	0.884	0.969	0.881
4	2S-Freq	6.054	4.459	3.279	2.429
	2S-Logit	1.146	1.176	1.043	1.130
	NPL	1.143	1.250	1.037	1.210
	2S-True	6.591	5.589	6.072	3.487
5	2S-Freq	1.162	1.209	1.020	1.166
	2S-Logit	1.158	1.248	1.010	1.197
	NPL	1.158	1.248	1.010	1.197
	2S-True	1.158	1.248	1.010	1.197

- The paper’s practical message comes directly from these tables: **do not trust raw two-step frequency-based PML in finite samples when the state space is large**; iterate with NPL instead.

5.3 Table VI: descriptive statistics for Chilean local retail markets

Read:

- Restaurants are far more numerous than the other retailers: about **14.6 firms per 10,000 people** versus roughly **0.7 to 1.9** in the other industries.
- Restaurants also have lower concentration and much smaller revenue per firm, which already suggests lower scale economies and perhaps weaker head-to-head strategic interaction.
- Entry and exit rates are high in all industries, and survival beyond the first few years is difficult. Gas stations have the strongest short-run survival among the five industries.
- This table motivates the need for a dynamic model with sunk costs and turnover rather than a static entry model.

5.4 Table VII: estimated number of potential entrants

Read:

- The paper treats the number of potential entrants as an estimable object that can vary across market-industry pairs.
- Most markets are estimated to have a small number of potential entrants, but the upper tail can be large, especially for restaurants.
- This table matters because the dynamics of entry and exit depend not only on current incumbents but also on the pool of firms that could enter.

5.5 Table VIII and Table IX: structural estimates and normalized interpretation

Read:

TABLE VI
DESCRIPTIVE STATISTICS: 189 MARKETS; YEARS 1994–1999

Descriptive Statistics	Restaurants	Gas Stations	Bookstores	Shoe Shops	Fish Shops
Number of firms per 10,000 people	14.6	1.0	1.9	0.9	0.7
Markets with					
0 firms	32.2%	58.6%	49.5%	67.1%	74.1%
1 firm	1.3%	15.3%	15.8%	10.8%	9.6%
2 firms	1.2%	7.8%	8.0%	6.7%	5.0%
3 firms	0.5%	5.2%	6.9%	3.8%	3.4%
4 firms	1.2%	4.0%	3.6%	2.7%	2.0%
More than 4 firms	63.5%	9.2%	16.2%	8.9%	5.9%
Herfindahl index (median)	0.169	0.738	0.663	0.702	0.725
Annual revenue per firm (in thousand \$)	17.6	67.7	23.3	67.2	124.8
Regression log(1+ # firms) on log(market size) ^a	0.383 (0.043)	0.133 (0.019)	0.127 (0.024)	0.073 (0.020)	0.062 (0.018)
Regression log(firm size) on log(market size) ^b	-0.019 (0.034)	0.153 (0.082)	-0.066 (0.050)	0.223 (0.081)	0.097 (0.111)
Entry rate (%) ^c	9.8	14.6	19.7	12.8	21.3
Exit rate (%) ^d	9.9	7.4	13.5	10.4	14.5
Survival rate (hazard rate)					
1 year (%) ^e	86.2 (13.8)	89.5 (10.5)	84.0 (16.0)	86.8 (13.2)	79.7 (20.3)
2 years (%)	69.5 (19.5)	88.5 (1.1)	70.0 (16.6)	71.1 (18.2)	58.1 (27.2)
3 years (%)	60.1 (14.9)	84.6 (4.3)	60.0 (14.3)	52.6 (25.1)	44.6 (23.3)

^aMarket size = population. Regression included time dummies. Standard errors are given in parentheses.
^bFirm size = revenue per firm. Regression included time dummies. Standard errors are given in parentheses.
^cEntry rate = entrants / incumbents.
^dExit rate = exits / incumbents.
^eSurvival and hazard rates are calculated using the subsample of new entrants in years 1995 and 1996.

TABLE VII
DISTRIBUTION OF THE ESTIMATED NUMBER OF POTENTIAL ENTRANTS

N_m	Restaurants	Gas Stations	Bookstores	Shoe Shops	Fish Shops
2	63 (33.3%)	146 (77.3%)	123 (65.1%)	153 (81.0%)	158 (83.6%)
3	1 (0.5%)	9 (4.8%)	14 (7.4%)	6 (3.2%)	6 (3.2%)
4	3 (1.6%)	8 (4.2%)	10 (5.3%)	8 (4.2%)	9 (4.8%)
5	1 (0.5%)	8 (4.2%)	5 (2.7%)	5 (2.7%)	2 (1.1%)
6	1 (0.5%)	3 (1.6%)	5 (2.7%)	4 (2.1%)	4 (2.1%)
Maximum	105	17	48	16	20

- Table VIII reports the raw NPL estimates. The fit is respectable even in this parsimonious model: the pseudo- R^2 measures for entry and exit are all economically meaningful.
- Table IX translates the estimates into economically interpretable ratios. Fixed operating costs are large—between about **59% and 85%** of the variable profit of a monopolist in a median-size market. Sunk entry costs are also important but generally smaller than annual fixed costs.
- Strategic interactions are statistically significant but modest in normalized terms: moving from monopoly to duopoly lowers variable profits by only about **6% to 11%** in a median market. This is particularly weak for restaurants and bookstores.
- Unobserved market heterogeneity is quantitatively important, explaining about **28% to 64%** of the cross-market variance in monopoly profits depending on the industry.
- The cross-industry interpretation is clear: restaurants have relatively low scale economies and weak strategic effects; gas stations have the largest sunk costs; bookstores have high fixed costs but weak strategic competition; fish shops display especially important unobserved market heterogeneity.

TABLE VIII
NPL ESTIMATION OF ENTRY–EXIT MODEL^a

Parameters	Restaurants	Gas Stations	Bookstores	Shoe Shops	Fish Shops
Variable profit:					
θ_{RS}	1.743	1.929	2.029	2.030	0.914
σ_ϵ	(0.045)	(0.127)	(0.076)	(0.121)	(0.125)
θ_{RN}	1.643	2.818	1.606	2.724	1.395
σ_ϵ	(0.176)	(0.325)	(0.201)	(0.316)	(0.234)
Fixed operating cost:					
θ_{FC}	9.519	12.769	15.997	14.497	6.270
σ_ϵ	(0.478)	(1.251)	(0.141)	(1.206)	(1.233)
Entry cost:					
θ_{EC}	5.756	10.441	5.620	5.839	4.586
σ_ϵ	(0.030)	(0.150)	(0.081)	(0.145)	(0.121)
σ_w	1.322	2.028	1.335	2.060	1.880
σ_ϵ	(0.471)	(1.247)	(0.133)	(1.197)	(1.231)
Number of observations	945	945	945	945	945
R-squared:					
Entries	0.298	0.196	0.442	0.386	0.363
Exits	0.414	0.218	0.234	0.221	0.298

^aStandard errors are given in parentheses. These standard errors are computed from the formulae in Section 4, which do not account for the error in the estimation of the parameters in the autoregressive process of market size.

TABLE IX
NORMALIZED PARAMETERS

Parameters ^a	Restaurants	Gas Stations	Bookstores	Shoe Shops	Fish Shops
$\frac{\theta_{FC}}{\theta_{RS} \ln(S_{Med})}$	0.590	0.716	0.852	0.772	0.742
$\frac{\theta_{EC}}{\theta_{RS} \ln(S_{Med})}$	0.357	0.585	0.299	0.311	0.542
$100 \frac{\theta_{RN} \ln(2)}{\theta_{RS} \ln(S_{Med})}$	7.1%	10.9%	5.9%	10.1%	11.4%
$\frac{\sigma_\epsilon^2}{\theta_{RS}^2 \text{var}(\ln(S)) + \sigma_w^2 + 1}$	0.278	0.436	0.235	0.423	0.642

^a $\theta_{FC}/(\theta_{RS} \ln(S_{Med}))$ is the ratio between fixed operating costs and variable profits of a monopolist in a market of median size. $S_{Med} = 10,400$ individuals. $\theta_{EC}/(\theta_{RS} \ln(S_{Med}))$ is the ratio between entry costs and variable profits of a monopolist in a market of median size. $100\theta_{RN} \ln(2)/(\theta_{RS} \ln(S_{Med}))$ is the percentage reduction in variable profits per firm when we go from a monopoly to a duopoly in a market of median size. $\sigma_\epsilon^2/(\theta_{RS}^2 \text{var}(\ln(S)) + \sigma_w^2 + 1)$ is the proportion of the cross-sectional variability in monopoly profits that is explained by the unobserved market type ω_m . Note that $\text{var}(\ln(S)) = 1.16$.

6 Short summary

- **Conclusion.** The paper makes estimation of dynamic discrete games feasible in environments with multiple equilibria and large state spaces by working with best-response probabilities rather than solving the full game inside every likelihood evaluation.
- **Main identification insight.** Structural parameters are identified from equilibrium conditional choice probabilities together with a representation of equilibrium as a fixed point of a computationally convenient mapping.
- **Main econometric lesson.** Two-step PML is fast but fragile in finite samples. NPL is the practical improvement because it reduces bias, can start from poor initial probability guesses, and extends naturally to settings with permanent unobserved heterogeneity.
- **Main empirical lesson.** In the Chilean retail application, dynamic entry and exit are shaped by large fixed costs, nontrivial sunk costs, moderate strategic interaction, and important unobserved market heterogeneity.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Dynamic discrete games can be estimated without imposing equilibrium uniqueness for every parameter vector and without solving every equilibrium by brute force.
- **Methodological contribution.** The paper builds a class of pseudo maximum likelihood estimators around a fixed-point representation of Markov perfect equilibrium in probability space.
- **Practical contribution.** The nested pseudo likelihood estimator is a tractable, robust workhorse that performs well in finite samples and can be used when the first-step nonparametric estimate of choice probabilities is weak or unavailable.

7.2 Economic intuition

- In a dynamic game, the hard part is not writing down payoffs; it is dealing with strategic beliefs and continuation values when many equilibria may exist.
- The paper’s solution is to view players’ conditional choice probabilities as beliefs, then repeatedly ask: given those beliefs, what is each player’s best response in probability space?
- This turns the estimation problem into something that looks much closer to a standard pseudo-likelihood problem, but with an equilibrium-consistency update built in.
- NPL works because each iteration disciplines noisy first-step beliefs using the model’s own best-response structure.

7.3 Takeaway

This paper is best understood as a bridge between dynamic structural IO and practical econometrics. Its key message is that dynamic games need not be estimated by fully solving every equilibrium for

every trial parameter vector. By recasting equilibrium in terms of best-response choice probabilities, the paper makes estimation computationally feasible and empirically useful. For applied work, the bottom line is simple: two-step PML is an attractive first pass, but NPL is the method that makes the framework genuinely practical.